

Yunxi Lai

Personal website: <https://yunxilai.github.io/> E-mail: laiyx29@mail2.sysu.edu.cn

Current affiliation: Zhongshan Ophthalmic Center, Sun Yat-sen University, China

EDUCATION

Ophthalmology | Zhongshan Ophthalmic Center, Sun Yat-sen University, China

Jul 2023–Jun 2026 **Doctor of Medicine (M.D.)** | Degree conferral expected December 2026

Thesis: Transformation from Clinical Data to Scientific Evidence Enabled by Large Language Models in Ophthalmology.

Sep 2020–Jun 2023 **Master of Medicine (M.M.)** | Licensed Physician

Standardized Residency Training Certificate in Ophthalmology

Thesis: Analyses of Routine Preoperative Blood Testing in Ocular Patients.

Clinical Medicine | Guangdong Medical University, China

Sep 2015–Jun 2020 **Bachelor of Medicine**

RESEARCH EXPERIENCE

Generative AI

Development, Validation and Implementation of an Ophthalmic Large Language Model

Jul 2023–Jun 2026

-Defined ophthalmic evaluation benchmarks for adapting a general-purpose LLM to ophthalmology and assessing its question-answering performance; co-authored an article published in *JAMA Ophthalmology* in 2024.

-Conceived its clinical application as a pre-consultation tool, designed an agent to actively elicit and document medical histories, and integrated it into existing outpatient workflows and hospital systems.

-Led full validation in a randomized controlled trial, including protocol and statistical analysis plan, ethics approval and trial registration, Python coding, statistical analysis in R and other trial operations; co-first-authored an article accepted by *npj Digital Medicine* on 30 August 2026.

-Coordinated multidisciplinary collaboration and dissemination across grant applications, intellectual property and public outreach, contributing to app deployment serving over 1.7 million people across four countries and to the project's selection as a Chinese digital health case study at the 2025 World Health Assembly.

Epidemiology

Routine Preoperative Blood Testing in Ophthalmic Surgery

Sep 2021–Jun 2023

- Independently conducted a retrospective study evaluating the necessity of routine preoperative blood testing and the applicability of *Choosing Wisely* recommendations to ophthalmic surgery in China.

- Analyzed and interpreted data from 102,542 patients, resulting in a first-author publication in the *British Journal of Ophthalmology* in 2024.

Cohort Studies	Pediatric Cohort on Ocular Effects of Systemic Medications
Aug 2023–Mar 2026	- Coordinated a cross-institutional ophthalmology–pediatrics collaboration to monitor ocular adverse effects associated with systemic medications, spanning ethics approval, participant recruitment, ophthalmic examinations, and longitudinal data management. - Co-established and managed a prospective cohort of 600+ children, with an average of seven follow-up visits per participant. - Designed an LLM-assisted extraction method with active data-quality control to address inefficient longitudinal data extraction, converting discharge summaries into structured research variables; internally evaluated using retrospective longitudinal inpatient records.
Nov 2023–Mar 2025	Community-Based Ophthalmic Cohorts - Explained test results, offered medical advice and collaborated with local healthcare providers.

TRANSFERABLE SKILLS

Clinical Visual Assessment: Experienced in comprehensive ophthalmic examination and imaging, including slit-lamp biomicroscopy, ophthalmoscopy, optical coherence tomography and scanning laser ophthalmoscopy; fine visuomotor coordination and procedural precision; demonstrated adaptability and commitment to high-quality work through surgical practice and rotations in emergency, inpatient and outpatient settings.

Research Design: Designed and conducted human-subject studies, including retrospective analyses, randomized controlled trials and prospective cohorts, from protocol development and registration to statistical analysis planning, data collection and analysis.

Quantitative and Computational Skills: Python and R programming for data processing, quantitative analysis and computational research; SPSS for statistical analysis; AI-assisted coding and prototyping using Codex and Visual Studio Code.

Project Management and Interdisciplinary Collaboration: Coordinated timelines and deliverables across clinical, technical and industry teams; aligned responsibilities and expectations.

Research Presentation: Drafted, revised and published two first-author manuscripts in peer-reviewed journals; gave and received constructive scientific feedback.

Languages: Chinese (native), English (fluent).

TEACHING & OUTREACH

Instructor	Pre-Rotation Training for First-Year Ophthalmology Residents
Aug 2024	- Prepared teaching materials and taught essential clinical procedures.
Contributor	Community Engagement & Outreach
2020-2025	- Led visual identity and art direction for the official launch of the ophthalmic LLM; contributed to video planning, content development and media promotion. - Planned and contributed to community activities, including Physicians' Day events, cultural performances and public eye-health screening initiatives.

CONFERENCE PRESENTATION

May 2025 **Evaluating GPT-4o in ophthalmic consultation and documentation: Bridging knowledge with clinical practice.** (Poster) *Association for Research in Vision and Ophthalmology (ARVO) Annual Meeting, Salt Lake City, Utah, USA.*

PUBLICATIONS

Peer-reviewed Journal Articles: 7; First-author: 2

1. Luo, M. J.*, **Lai, Y.***, Wang, W.*, Bi, S., Jin, L., Chen, J., Wu, X., Lin, Z., Wang, J., Pang, J., Dongye, M., Wu, Y., Liu, Z., & Lin, H. (Accepted August 30, 2026. *Equal contribution) Standardized Pre-consultation by a Large Language Model Agent vs Ophthalmology Residents: A Randomized Clinical Trial. *npj Digital Medicine*.
2. **Lai, Y.**, Zeng, W., Liao, J., Yu, Y., Liu, X., & Wu, K. (2024) Retrospective Analyses of Routine Preoperative Blood Testing in a Tertiary Eye Hospital: Could Choosing Wisely Work in China? *British Journal of Ophthalmology*, 108(6), 897–902.
3. Luo, M. J., Pang, J., Bi, S., **Lai, Y.**, Zhao, J., Shang, Y., Cui, T., Yang, Y., Lin, Z., Zhao, L., Wu, X., Lin, D., Chen, J., & Lin, H. (2024) Development and Evaluation of a Retrieval-Augmented Large Language Model Framework for Ophthalmology. *JAMA Ophthalmology*, 142(9), 798–805.
4. Luo, M. J., Bi, S., Pang, J., Liu, L., Tsui, C. K., **Lai, Y.**, Chen, W., Yang, Y., Xu, K., Zhao, L., Jin, L., Lin, D., Wu, X., Chen, J., Chen, R., Liu, Z., Zou, Y., Yang, Y., Li, Y., & Lin, H. (2025) A Large Language Model Digital Patient System Enhances Ophthalmology History Taking Skills. *npj Digital Medicine*, 8(1), 502.
5. Liu, L., Hong, J., Wu, Y., Liu, S., Wang, K., Li, M., Zhao, L., Liu, Z., Li, L., Cui, T., Tsui, C. K., Xu, F., Hu, W., Yun, D., Chen, X., Shang, Y., Bi, S., Wei, X., **Lai, Y.**, Lin, D., Fu, Z., Deng, Y., Cai, K., Xie, Y., Cao, Z., Wang, D., Zhang, X., Dongye, M., Lin, H., & Wu, X. (2024) Digital ray: Enhancing Cataractous Fundus Images Using Style Transfer Generative Adversarial Networks to Improve Retinopathy Detection. *British Journal of Ophthalmology*, 108(10), 1423–1429.
6. Shang, Y., Lin, Y., Li, R., Shang, Y., Li, M., Zhao, L., Jin, L., Xu, A., Liu, D., Wu, Q., Luo, M., Pang, J., Bi, S., He, Y., Xu, M., Chen, X., Cao, Z., Yu, S., Zhao, J., **Lai, Y.**, Chen, W., & Lin, H. (2025) The Effectiveness of Large Language Models in Medical AI Research for Physicians: A Randomized Controlled Trial. *Cell Reports Medicine*, 6(12), 102469.
7. Li, Y., Cui, T., Lin, Z., Zhao, L., **Lai, Y.**, Yun, D., Lin, Z., Li, M., Huang, Y., Chen, S., Wang, D., Qiu, Y., Xu, Y., Yang, Y., Lin, D., Jiang, X., & Lin, H. (2026) Risk Factors and Critical Thresholds for Glucocorticoid-Induced Ocular Adverse Reactions in Children: Implications for Muscle and Neuromuscular Disease Management. *European Journal of Translational Myology*, 36(2).